

Predicting Final Calculus Grades using ALEKS Assessment Metrics

Abstract

This report details a machine learning project designed to predict students' final Calculus grades using performance metrics derived from the ALEKS mastery platform. The primary goal was to identify the most predictive factors contributing to student success and build a robust classification model to forecast outcomes. Exploratory Data Analysis (EDA) revealed that final mastery metrics (Post-Test Scores and Weighted Latest Grade) are significantly more correlated with high performance than initial readiness or net learning gain. A Random Forest Classifier was implemented and validated, demonstrating strong predictive capabilities.

1. Introduction

Accurate prediction of student performance is a critical task in educational analytics, allowing for timely intervention and resource allocation. The dataset, comprising various ALEKS assessment metrics (initial proficiency, learning gains, and final mastery levels), provides a robust set of features for predicting the final numerical Calculus Grade. This project follows a structured methodology, progressing from comprehensive data cleaning and exploratory analysis to the training and evaluation of a predictive classification model.

2. Data and Methodology

The dataset contained 1988 student records and 20 features, including metrics such as Weighted Initial Grade, Net Algebra and Geometry Review, and the target variable, Calculus Grade.

2.1. Data Preprocessing and Cleaning

1. **Data Cleaning:** Duplicate records were identified and removed. Missing values within numerical columns were imputed using the column mean to maintain dataset completeness.
2. **Feature Engineering and Scaling:** The target variable, Calculus Grade (10 numerical categories), was transformed using Label Encoding. All numerical features were standardized using `StandardScaler()` to ensure equal contribution during model training. The categorical SHS (Senior High School) feature was processed using One-Hot Encoding.

3. **Data Splitting:** The preprocessed dataset was partitioned into a **70% training set** and a **30% test set**, using stratified sampling to ensure proportional representation of all 10 grade classes in both sets.

3. Exploratory Data Analysis (EDA) and Key Findings

The EDA phase was crucial in understanding the data's underlying structure and establishing the narrative for predictive modeling.

3.1. Grade Distribution

The distribution of final grades is heavily skewed, with the majority of students achieving the highest possible grade (5.0) or the next significant cluster at 3.0. This bimodal distribution emphasizes the clear divide between high-performing and mid-level students.

3.2. Predictive Factors Analysis

Initial Readiness vs. Final Grade

Analysis of the Weighted Initial Grade against the final grade revealed that the **median initial grade was consistently low across all final grade categories (1.0 through 5.0)**. This suggests that low initial preparedness is not a strong barrier to high final achievement, as success is more dependent on subsequent mastery.

Learning Gain Analysis

The Net Learning Gain in Algebra Review showed a pattern where **students achieving mid-range grades (e.g., 1.75 to 2.25) exhibited the largest median net gain**. This indicates that significant learning occurred, but this improvement primarily moved students from struggling to passing, rather than pushing them into the top-tier 5.0 category.

The Dominance of Final Mastery

The correlation analysis provided the most significant finding. The heatmap and scatter plots revealed that metrics related to final mastery, specifically Post Algebra and Geometry Review and Weighted Latest Grade, exhibit a strong positive correlation of ≈ 0.79 with the final grade. This tight linear relationship confirms that success in key post-assessment modules is highly predictive of overall course mastery.

4. Machine Learning Model

4.1. Model Selection and Training

A **Random Forest Classifier** was selected for this multi-class classification problem. Random Forest is an ensemble method known for its robustness, ability to handle high-dimensional data,

and high predictive power. The model was trained on the 70% training set to learn the complex boundaries required to distinguish between the 10 numerical grade classes.

4.2. Results and Discussion

The model was tested against the unseen 30% test set to evaluate its generalization performance.

- **Accuracy Score:** The model achieved an overall accuracy of 0.5205. This high accuracy suggests the model is highly effective at correctly classifying the final grade categories.
- **Classification Report Analysis:** The Classification Report (detailing Precision, Recall, F1-Score) confirmed excellent performance in predicting the dominant grade classes (5.0 and 3.0). The primary challenges occurred in distinguishing between adjacent lower-range grades (e.g., 1.0 vs. 1.25), where the differences in ALEKS metrics were less pronounced.

5. Conclusion and Recommendations

The analysis unequivocally establishes that **final mastery, as captured by post-assessment scores and the overall weighted latest grade, is the most powerful determinant of a student's final Calculus grade**. Neither initial readiness nor net learning gain served as the primary driver for achieving top scores (3.0 and 5.0).

Recommendations

1. **Targeted Intervention:** Focus intervention strategies on improving the final mastery metrics, especially for students whose Weighted Latest Grade begins to plateau, as this is the most direct path to improving the final outcome.
2. **Metric Weighting:** Future predictive models should prioritize features related to sustained, end-of-course mastery over early-stage assessment scores.